

Cognizant Deepskilling Modules

Engineering Concepts:

Design Patterns and Principles:

Exercise 1: Singleton Pattern – Output

```
C:\Program Files\Java\jdk-24\bin\java.exe "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60828" -Dfile.encoding=UTF-8
Logger Instance Created
LOG: Application Started
LOG: User Logged In

Are both objects same? true

Process finished with exit code 0
```

Exercise 2: Factory Method – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=64030" -Dfile.encoding=UTF-8
Opening Word Document
Opening PDF Document
Opening Excel Document

Process finished with exit code 0
```

Exercise 3: Builder Pattern – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=64012" -Dfile.encoding=UTF-8
Computer Configuration
-----
CPU : Intel i7
RAM : 16 GB
Storage : 512 GB SSD
Graphics Card : NVIDIA RTX 4060
Operating System : Windows 11

Process finished with exit code 0
```

Exercise 4: Adapter Pattern – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=64023" -Dfile.encoding=UTF-8
Payment of ₹2500.0 made using PayPal.
Payment of ₹5000.0 made using Stripe.

Process finished with exit code 0
```

Exercise 5: Decorator Pattern – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60868" -Dfile.encoding=l
----- Email Only -----
Email Notification: Order Confirmed

----- Email + SMS -----
Email Notification: Order Confirmed
SMS Notification: Order Confirmed

----- Email + SMS + Slack -----
Email Notification: Order Confirmed
SMS Notification: Order Confirmed
Slack Notification: Order Confirmed

Process finished with exit code 0

```

Exercise 6: Proxy Pattern – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60870" -Dfile.encoding=l
Mobile App: TCS price updated to ₹4200.0
Web App: TCS price updated to ₹4200.0

Mobile App: Infosys price updated to ₹1650.0
Web App: Infosys price updated to ₹1650.0

Process finished with exit code 0

```

Exercise 7: Observer Pattern – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60870" -Dfile.encoding=l
Mobile App: TCS price updated to ₹4200.0
Web App: TCS price updated to ₹4200.0

Mobile App: Infosys price updated to ₹1650.0
Web App: Infosys price updated to ₹1650.0

Process finished with exit code 0

```

Exercise 8: Strategy Pattern – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=58682" -Dfile.encoding=l
Paid ₹2500.0 using Credit Card.
Paid ₹5000.0 using PayPal.

Process finished with exit code 0

```

Exercise 9: Command Pattern – Output

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60870" -Dfile.encoding=l
Light is ON
Light is OFF

Process finished with exit code 0

```

Exercise 10: MVC Pattern – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=60875" -Dfile.encoding=UTF-8
Student Details
-----
Name : Bindhu
ID : 101
Grade : A

Student Details
-----
Name : Ujwala
ID : 101
Grade : A+

Process finished with exit code 0
```

Exercise 11: Dependency Injection – Output

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\msree\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=58937" -Dfile.encoding=UTF-8
Customer ID : 101
Customer Name : Bindhu

Process finished with exit code 0
```

M.sree bindhu